

Resonances, Spectral Theory, and the Resurgent Structure

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- OM and S. Ogawa, PTEP [arXiv:2503.18741 [hep-th]].
- OM and S. Ogawa, JHEP [arXiv:2505.02301 [hep-th]].
- OM and S. Ogawa, arXiv:2508.09211 [quant-ph].

(Collaboration with nuclear physicist)

Spectral theory

- Finite-dim Hilbert space is well-known
 - ▶ Complete, metric, linear space
- Quantum theory addresses infinite-dim Hilbert space
- ① Discrete spectrum: $\langle E|E'\rangle = \frac{1}{\mathcal{N}^2}\delta_{E,E'}$
- ② Continuum spectrum: $\langle E|E'\rangle \propto \delta(E - E')$
 - ▶ Let φ_l be regular solution of radial Schrödinger equation
 - ★ φ_l is subject to

$$\varphi_l(k, r) \xrightarrow{r \rightarrow 0} j_l(kr) \quad j_l: \text{Spherical Bessel function}$$

with angular momentum l .

- ▶ At $r \rightarrow \infty$, linear combination of Hankel functions h_l :

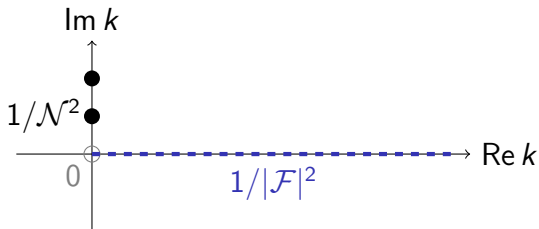
$$\varphi_l(k, r) \rightarrow \mathcal{F}_l(k)h_l^-(kr) - \mathcal{F}_l^*(k)h_l^+(kr) \quad \mathcal{F}_l: \text{Jost function}$$

$$\begin{array}{c} \text{blue wavy arrow} \rightarrow h^+ \\ \text{red wavy arrow} \rightarrow h^- \end{array}$$

● $\longrightarrow r$ Normalize in terms of $\mathcal{F}, \mathcal{F}^*$

Spectral function and completeness

- The weight function in inner prod, spectral function, distributes



- From integration of Green function as ,

$$\text{Res: } \underbrace{\sum_i |\varphi_i\rangle\langle\varphi_i|}_{\text{Bound states}} + \underbrace{\int_0^\infty dk |\varphi(k)\rangle\langle\varphi(k)|}_{\text{Continuum states}} = 1$$

- Quantum mechanics works well!

Advanced topic: Rigged Hilbert space

- How to treat unbounded operators
 - ▶ $\mathcal{H} \equiv \{\text{square-integrable wavefunctions}\}$
 - ▶ Only on a subdomain $D(A)$ on which A is self-adjoint.
- Set of operators $\{A_i\}$; then let us introduce the dense subspace

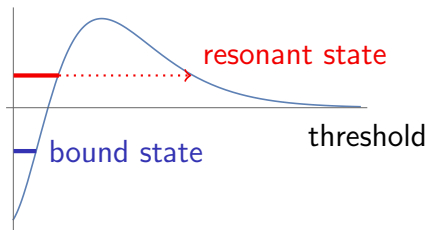
$$\Phi \equiv \cap_i D(A_i) \subseteq \mathcal{H}.$$

- $|x\rangle, |p\rangle$ require larger linear space, so antilinear functional: $\Phi^\times$
 - ▶ For $|x\rangle \in \Phi^\times$, $\varphi = \int dx |x\rangle \langle x| \varphi \rangle \in \Phi$.
 - ▶ Φ : Physical, $\Phi^\times$: Functional over Φ .
- We find the Gelfand multiplet (rigged Hilbert space)

$$\Phi \subseteq \mathcal{H} \subset \Phi^\times.$$

Quantum mechanical resonance

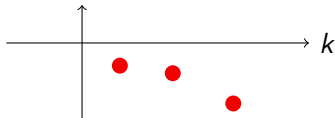
- (Usual) Quasi-stationary state in quantum system
 - ▶ There are some local minima (vacua); one decays to an other.
 - ▶ E.g., double well potential, periodic cosine well, ...
 - ▶ After decaying, finally stable ground state.
 - ▶ Eventually bound state **beyond perturbation theory**.
- Unstable state after decay: **Resonant state**



- ▶ Similar/related to scattering theory, nuclear physics, open system, and non-Hermitian QM

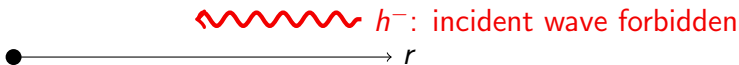
Problems of resonant states

- Simply say, plane wave in asymptotic region



- But, k is not in integral contour.
- Non-normalizable wavefunctions: $e^{ikr} = e^{ik_R r + k_I r} \rightarrow \infty$

- Moreover, at $r \rightarrow \infty$, Siegert boundary condition



implies $\mathcal{F} = 0 \rightarrow$ spectral function? (or ill-def resolvent $\frac{1}{H-z}$)

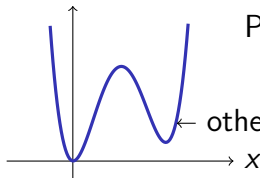
- Completeness?
 - $\sum_{\text{bound}} |\psi_B\rangle\langle\psi_B| + \int_{\text{scattering}} |\psi_S\rangle\langle\psi_S| \stackrel{?}{=} 1$, but $|\psi_R\rangle\langle\psi_R|$???
- What is the wavefunction itself?
 - Complex probability; expectation value?
 - Transition cross-section?

Subtleties of resonance physics

- Resonance appears quite universally!
- We aim to understand it in not only math but also phys senses
- There is a large gap between phenomenological success & math
 - ▶ **Complex scaling method**: $r \rightarrow re^{i\theta}$ for large enough θ .
 - ▶ Mathematically well-defined by Aguilar–Balslev–Combes (ABC) theorem [AC '71, BC '71] (as I will explain later).
 - ▶ Physics depends on θ explicitly, but computationally successful.
- Quite well-established definition/regularization (historically [> 50 years], experimentally, monumentally)
- To make it transparent, consider the **complex- x** plane
- Focus on analytic solutions of diff eq w.r.t. $x \in \mathbb{C}$

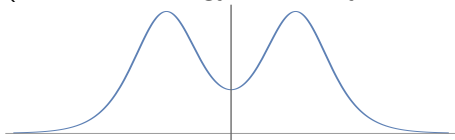
Resurgence approach to quasi-stationary states

- E.g., for double well potential, perturbation theory suffers from



Perturbative vac decays w/ decay rate Γ
→ Resurgent asymptotic series

- Complex energy of resonance: $E = E_r - i\frac{\Gamma}{2}$
(Resonant energy E_r , decay width Γ)



E.g., $V(x) = \frac{U_0}{\cosh^2 \beta(x-a)} + \frac{U_0}{\cosh^2 \beta(x+a)}$

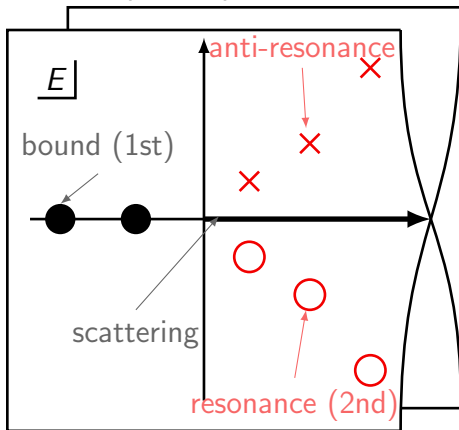
- Can resurgence theory overcome difficulties as **non-perturbative formulation of QM**?
- If so, *essence in “ $\forall$ QM” = analyticity in resurgence*
 - ▶ Quite transparent and precise!!!

Review I: Resonance as S-matrix pole

- S-matrix is written in terms of Jost function

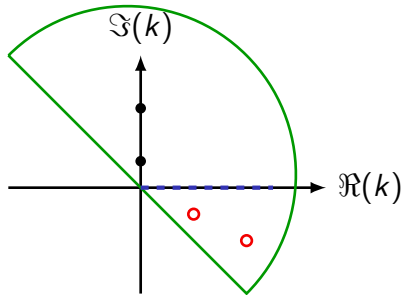
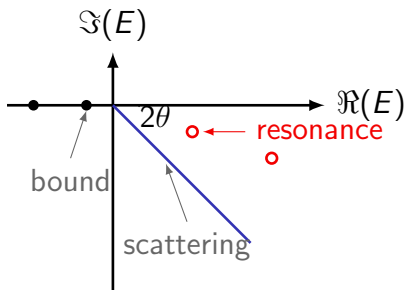
$$S_I(k) = \frac{\mathcal{F}_I^*(k)}{\mathcal{F}_I(k)} \rightarrow \infty \quad \text{pole singularity at resonance}$$

- Riemann surface of complex E -plane



Prescription of complex scaling

- Complex scaling method: Radial direction $r \rightarrow re^{i\theta}$ ($k \rightarrow ke^{-i\theta}$)



- ABC theorem [See Myo-Kikuchi-Masui-Katō '14]
 - Resonant wavefunctions are square-integrable.
 - Bound/resonant *energy* is independent from θ .
 - Continuous spectrum is rotated by 2θ .
- Deformation of integral contour of Green func.
 - It includes bound, continuous, and resonant states.
 - $\sum_B |\psi_B^\theta\rangle \langle \tilde{\psi}_B^\theta| + \sum_{R_\theta} |\psi_R^\theta\rangle \langle \tilde{\psi}_R^\theta| + \int_{S_\theta} |\psi_S^\theta\rangle \langle \tilde{\psi}_S^\theta| = 1$ (not *-ring?)

Review II: Complexification as non-pert. method

- Rough sketch:

Perturbation theory in quasi-stationary system provides asymptotic series growing factorially, $\sum_n n! \lambda^n$.

- 1 Computation with complex coupling $\lambda \rightarrow e^{\pm i\epsilon} \lambda$
 - ▶ Resummation technique $\rightarrow$ convergent if $\epsilon > 0$
 - ▶ Ambiguity with $\epsilon \rightarrow 0 = \pm iO(e^{-1/\lambda})$: non-perturbative
 - 2 Steepest descent around saddle point $x \in \mathbb{C}$ (Lefschetz thimble)
 - ▶ $\underbrace{(\text{pert.})}_{\text{finite} \pm \text{ambiguity}} \mp \underbrace{e^{-1/\lambda}}_{\text{nontrivial saddle}} (\text{pert.}) + \dots$
- Resurgent (no ambiguity)
 - Available toolkit here: Exact WKB analysis [Voros]

More issue: inverted Rosen–Morse potential

- Potential giving resonance

$$V(x) = \frac{U_0}{\cosh^2 \beta(x-a)} + \frac{U_0}{\cosh^2 \beta(x+a)}$$

↓

$$V(x) = U_0 / \cosh^2 \beta x$$

- Exact solution of Schrödinger equation is given by

$$\psi(x) = (1 - \xi^2)^{-\frac{ik}{2\beta}} F\left(-\frac{ik}{\beta} - s, -\frac{ik}{\beta} + s + 1, -\frac{ik}{\beta} + 1, \frac{1 - \xi}{2}\right)$$

F : Gauss hypergeometric function, $\xi = \tanh \beta x$, $k = \sqrt{2mE}/\hbar$,

$$s = \frac{1}{2}(-1 + \sqrt{1 - 8mU_0/\beta^2\hbar^2})$$

- Resonant energy

$$E_n^R = \frac{\hbar^2 \beta^2}{8m} \left[\sqrt{\frac{8mU_0}{\beta^2 \hbar^2}} - 1 - i(2n + 1) \right]^2$$

- **Barrier resonance**: localized state at potential bump (?)
 - ▶ Not intuitively clear why this is happening.

WKB ansatz

- Consider a Schrödinger equation

$$\left(-\frac{d^2}{dx^2} + \hbar^{-2} Q(x) \right) \psi = 0, \quad Q(x) = 2[V(x) - E]$$

- Introduce WKB ansatz as a formal power series

$$\psi(x, \hbar) = e^{\int^x S(x', \hbar) dx'}, \quad S(x, \hbar) = \frac{1}{\hbar} S_{-1}(x) + S_0(x) + \hbar S_1(x) + \dots,$$

and substitute this into the equation; we have a recursive equation of S_i

- We have two solutions because of the leading-order equation

$$S_{-1}^2 = Q \quad \Rightarrow \quad S_{-1} = S_{-1}^{\pm} \equiv \pm \sqrt{Q}.$$

Solutions are $\psi^{\pm} = e^{\int S^{\pm}} \sim e^{\pm \frac{1}{\hbar} \int \sqrt{Q}}.$

Borel resummation

- Borel resummation: summing divergent asymptotic series

$$f(\lambda) \sim \sum_{k=0}^{\infty} f_k \lambda^{k+1} \quad \text{with } f_k \sim a^k k! \text{ as } k \rightarrow \infty$$

⇓ Borel transform

$$B(u) \equiv \sum_{k=0}^{\infty} \frac{f_k}{k!} u^k = \frac{1}{1 - au} \quad (\text{Pole singularity at } u = 1/a).$$

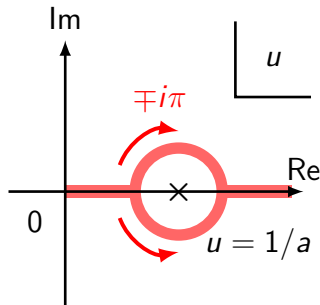
- The Borel sum is given by

$$f(\lambda) \equiv \int_0^{\infty} du B(u) e^{-u/\lambda}.$$

- $a < 0$ (alternating series) $\rightarrow$ convergent

- $a > 0 \rightarrow$ ill-defined due to the pole

$\Rightarrow$ Imaginary ambiguity $\sim \pm e^{-1/(a\lambda)}$

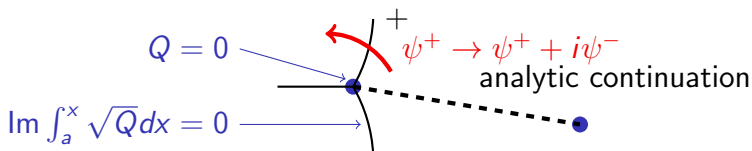


Stokes phenomena, Stokes geometry

- WKB ansatz is also asymptotic series; where is it Borel summable?
- Integral contour includes $\mp \int^x dx' \sqrt{Q}$ for $\psi^\pm$

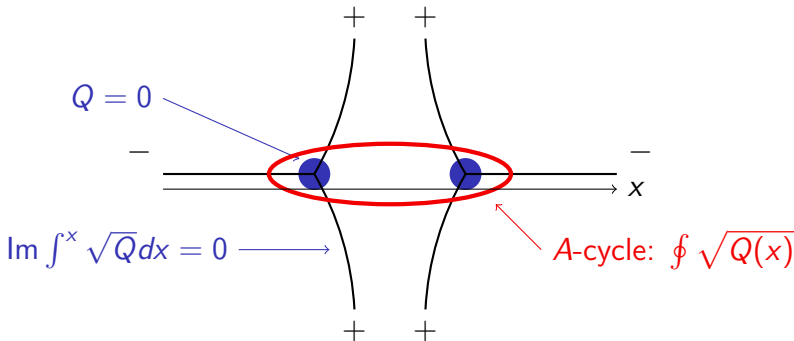
$$\text{Stokes curve : } \left\{ x \left| \operatorname{Im} \int_a^x dx' \sqrt{Q} = 0, \text{ where } Q(x=a) = 0 \right. \right\}$$

- ▶ a : turning point.
- ▶ Not on Stokes curve, Borel summable (analytically continuable).
- ▶ Across a Stokes curve, solution suddenly changes.
- ▶ $\psi^\pm$ is dominant with $\operatorname{Re} \int_a^x dx' \sqrt{Q} \gtrless 0$.



Quantization condition from normalizability

- (Usual) QM: normalizable from $x \rightarrow -\infty$ to $x \rightarrow \infty$
 - ▶ E.g., Harmonic oscillator

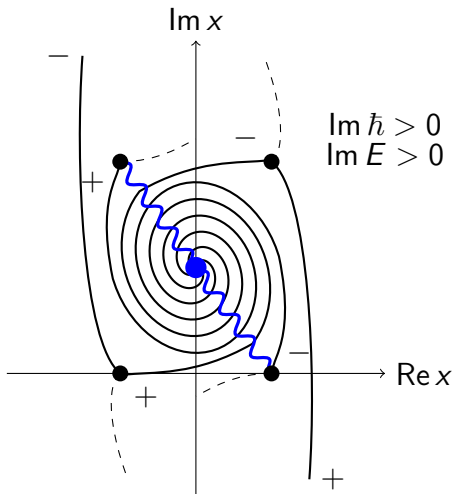
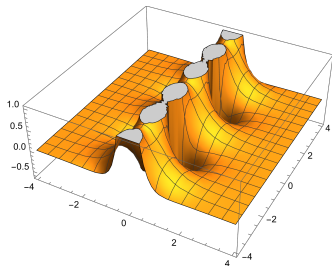
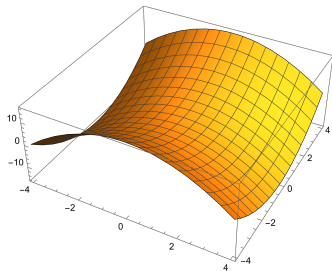


- ▶ $\psi_{\text{final}} = (\text{analytic continuation, Stokes pheno}) \times \psi_{\text{initial}}$
- ▶ Non-normalizable solution vanishes: $\psi_{\text{initial}}^{\text{damp}} \not\rightarrow \psi_{\text{final}}^{\text{grow}}$

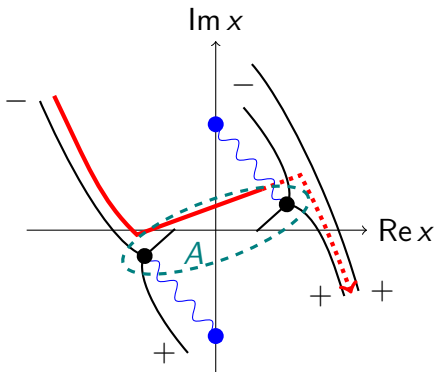
$$1 + A = 1 + e^{\hbar^{-1} \oint \sqrt{Q}} = 0 \quad \Rightarrow \quad E = \hbar \left(n + \frac{1}{2} \right)$$

How to see resonant state from exact WKB

- Resonant state: quasi-stable in complex region
 - ▶ $V = -x^2$ vs $V = 1/\cosh^2 x$



Leading order estimate (Rigorous one in next slide)



- Normalizable path:
- Barrier resonant energy from exact WKB analysis

► A-cycle

$$\int_{-\cosh^{-1} \sqrt{\frac{1}{E}}}^{\cosh^{-1} \sqrt{\frac{1}{E}}} \sqrt{2 \left(\frac{1}{\cosh^2 x} - E \right)} = \sqrt{2}\pi(1 + \sqrt{E})$$

quantization condition: $1 - A = 0$ $E = \left(1 - \frac{in}{\sqrt{2}}\right)^2$

Rigorous estimate being identical to exact solution

- $A = 1$:

$$\left[\frac{F\left(-\frac{ik}{\beta} - s, -\frac{ik}{\beta} + s + 1, -\frac{ik}{\beta} + 1, \frac{1-|\xi|}{2}\right)}{F\left(-\frac{ik}{\beta} - s, -\frac{ik}{\beta} + s + 1, -\frac{ik}{\beta} + 1, \frac{1+|\xi|}{2}\right)} \right]^2 = 1,$$

where $|\xi| = \tanh(\beta \cosh^{-1} \sqrt{U_0/E})$.

- Here, we use the formula of the hypergeometric function as

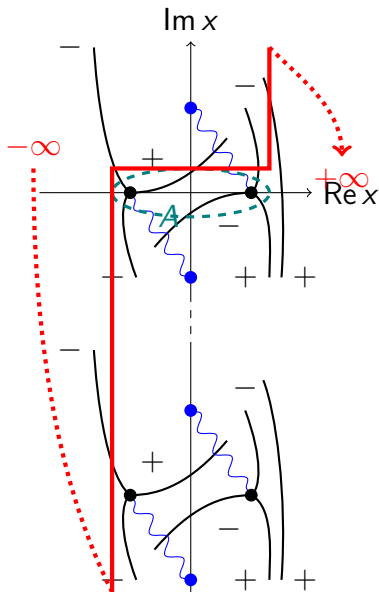
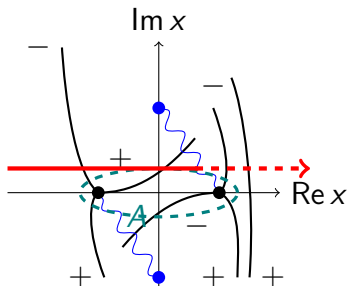
$$\begin{aligned} & F\left(-\frac{ik}{\beta} - s, -\frac{ik}{\beta} + s + 1, -\frac{ik}{\beta} + 1, \frac{1 \pm |\xi|}{2}\right) \\ &= \mathfrak{A} F\left(-\frac{ik}{2\beta} - \frac{s}{2}, -\frac{ik}{2\beta} + \frac{s+1}{2}, \frac{1}{2}, |\xi|^2\right) \mp \mathfrak{B} F\left(-\frac{ik}{2\beta} - \frac{s-1}{2}, -\frac{ik}{2\beta} + \frac{s}{2} + 1, \frac{3}{2}, |\xi|^2\right), \end{aligned}$$

where

$$\mathfrak{A} = \frac{\Gamma\left(-\frac{ik}{\beta} + 1\right) \Gamma\left(\frac{1}{2}\right)}{\Gamma\left(-\frac{ik}{2\beta} - \frac{s-1}{2}\right) \Gamma\left(-\frac{ik}{2\beta} + \frac{s}{2} + 1\right)}, \quad \mathfrak{B} = \frac{\Gamma\left(-\frac{ik}{\beta} + 1\right) \Gamma\left(-\frac{1}{2}\right)}{\Gamma\left(-\frac{ik}{2\beta} - \frac{s}{2}\right) \Gamma\left(-\frac{ik}{2\beta} + \frac{s+1}{2}\right)}.$$

- $\mathfrak{A} = 0, \mathfrak{B} = 0$: odd/even number of nodes

What is problematic in naive path? [1/3]



What is problematic in naive path? [2/3]

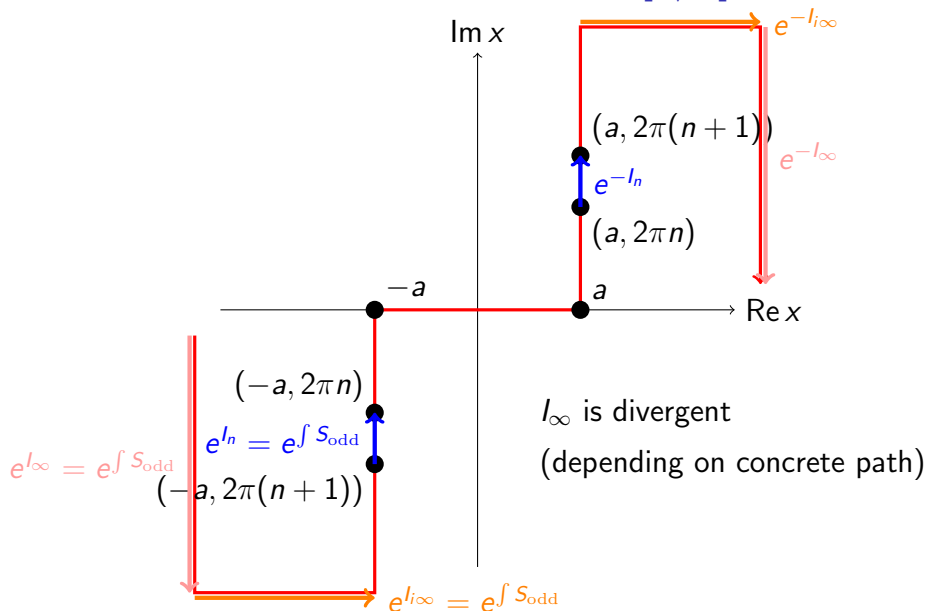
- Total quantization condition is given by

$$\begin{aligned} 0 &= e^{\int_{a+i\infty}^{a+\infty} dx S_{\text{odd}}} \left[\prod_{n=0}^{\infty} e^{\int_{a+2\pi i n/\beta}^{a+2\pi i(n+1)/\beta} dx S_{\text{odd}}} \right] (1 - A) \\ &\quad \times \left[\prod_{n=0}^{\infty} e^{\int_{-a-2\pi i(n+1)/\beta}^{-a-2\pi i n/\beta} dx S_{\text{odd}}} \right] e^{\int_{-a-i\infty}^{-a-\infty} dx S_{\text{odd}}} \\ &\stackrel{?}{=} e^{\int_a^{-a} dx S_{\text{odd}}} (1 - A), \end{aligned}$$

► $S_{\text{odd}} = \frac{1}{\hbar} S_{-1} + \hbar S_1 + \dots$

- Naively, any factor at infinity appears to cancel against each other.

What is problematic in naive path? [3/3]



Zel'dovich regularization ['61, Berggren '68]

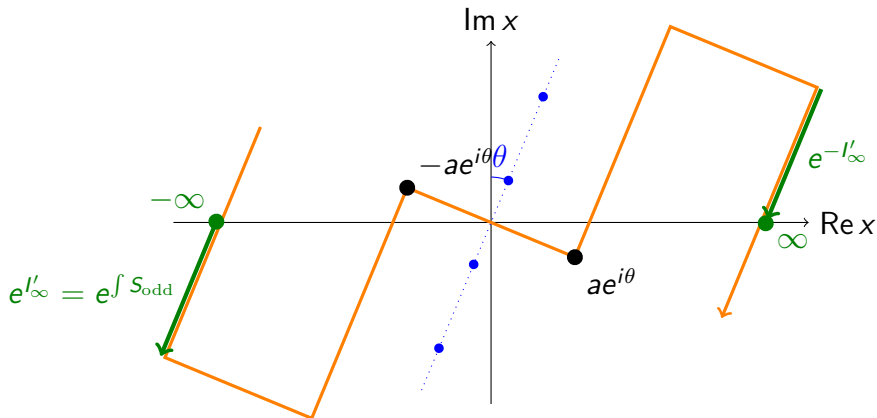
- A simple regularization
- Zel'dovich transformation: $S = e^{-\epsilon x^2}$

$$\begin{aligned} H\psi = E\psi &\quad \rightarrow \quad H_S\psi_S = E\psi_S, \\ \psi_S \equiv S\psi, \quad H_S &\equiv SHS^{-1}. \end{aligned}$$

- ▶ We find $H_S = H - \frac{\hbar^2\epsilon}{m} \left(\frac{d}{dx}x + x\frac{d}{dx} \right)$
- Now, $e^{I_\infty} = e^{\int (S_{\text{odd}} - 2\epsilon x)}$ becomes finite.

Complex scaling method

- Complex scaling by θ rotates the Stokes graph as



- Now, $e^{I_\infty} \rightarrow e^{I'_\infty}$ becomes finite, and A -cycle is unchanged.

ABC theorem

- ① Resonant wavefunctions are square-integrable.
 - ▶ Yes, if the path connects $\text{Im } x \pm \infty$ or is regularized. It straightforwardly leads to discrete and complex energies.
- ② Bound/resonant energy is independent from θ .
 - ▶ Potential and contour are rotated simultaneously; A is invariant.
- ③ Continuum spectrum is rotated by 2θ .
 - ▶ To keep Siegert b.c. (e^{ikx}), $k \rightarrow e^{-i\theta} k$ when $x \rightarrow e^{i\theta} x$.
 - ▶ Due to $k = \frac{\sqrt{2mE}}{\hbar}$, $E \rightarrow e^{-2i\theta} E$.

Rigged Hilbert space with resonance

- We introduce

$$\mathcal{D}_\varepsilon^R = \{x \in \mathbb{C} \mid \varepsilon > 0, \lim_{r \rightarrow \pm\infty} |x - r| < \varepsilon\}$$

- ▶ $\mathcal{D}_\varepsilon^R$ is the most crucial singular region.
- if $x \in \mathbb{C} \setminus \mathcal{D}_\varepsilon^R$, the Hilbert space is well-defined, $\mathcal{H}_\varepsilon$
 - ▶ Def. of norm and inner product is quite different!
 - ▶ Resonance is included as “bound” state.
- (Operator algebra) Set of operators $\{A_i\}$ where A_i is defined on $D(A_i) \subset \mathcal{H}_\varepsilon$; then let us introduce the dense subspace

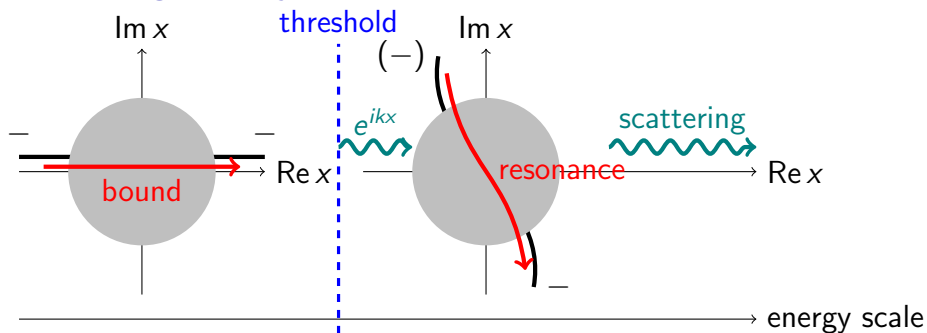
$$\Phi \equiv \cap_i D(A_i) \subseteq \mathcal{H}_\varepsilon.$$

- The range is defined by the limit as $\varepsilon \rightarrow 0$, say, $\Phi^\times$
- We find the Gelfand multiplet

$$\Phi \subseteq \mathcal{H}_\varepsilon \subset \Phi^\times,$$

and $(\mathcal{H}_\varepsilon, \Phi)$ is the rigged Hilbert space.

Scattering theory



- Below threshold: Bound states
- Above threshold: Resonance & continuum spectra
 - ▶ Read transmission/reflection coefficient from monodromy & analytic continuation
 - ▶ Transmission coefficient is divergent if E comes across resonance.

Summary

- We develop a **unified** framework for analyzing quantum mechanical **resonances** using **exact WKB method**
 - ▶ Serving as a non-perturbative formulation of QM.
 - ▶ Incorporating the Zel'dovich regularization, the complex scaling method, and the rigged Hilbert space.
 - ▶ Demonstrated by examining the inverted Rosen–Morse potential.
- Future works:
 - ▶ Realistic potential (E.g., $1/\cosh^2(x-a) + 1/\cosh^2(x+a)$)
 - ▶ Cross-section (Incident wave with connection&monodromy)
 - ▶ Numerical approach
 - ★ Quantization condition → Estimate cycles
 - ▶ Unitarily equivalence in Stone–von Neumann???
 - ▶ Higher dim???